Service Now System Administrator

Sub topic :Smart Library Workflow in ServiceNow

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1. INTRODUCTION

In today's digital era, educational institutions and organizations are increasingly adopting cloud-based platforms to automate their daily operations and improve service efficiency. Traditional library management systems often rely on manual processes for maintaining book records, issuing and returning books, tracking due dates, and managing member information. These manual activities are time-consuming, prone to human errors, and make it difficult to maintain accurate and up-to-date records. As the number of library users and resources continues to grow, there is a need for an intelligent and automated system that simplifies library operations while improving user experience.

ServiceNow is a leading cloud-based digital workflow platform that enables organizations to automate business processes through configurable applications, workflows, forms, notifications, and reporting features. It provides a low-code/no-code development environment that allows administrators to design business applications without extensive programming knowledge. ServiceNow System Administrators play a vital role in configuring the platform by creating tables, forms, roles, workflows, business rules, Flow Designer automations, notifications, and reports to streamline organizational processes.

The **Smart Library Workflow in ServiceNow** is developed to automate the complete library management process using the ServiceNow platform. The application allows students or members to request books through the Service Portal, while librarians manage approvals, issue books, record returns, and monitor inventory through a centralized system. The workflow automatically checks book availability, assigns due dates, updates book status, sends email notifications, and generates reports for effective library administration.

The project utilizes several core ServiceNow features, including custom tables, forms, Flow Designer, Business Rules, Client Scripts, UI Policies, Notifications, Roles, Reports, and Dashboards. By integrating these components, the system ensures secure access, efficient workflow execution, accurate record management, and real-time visibility into library operations.

The Smart Library Workflow offers numerous advantages to educational institutions, schools, colleges, universities, and corporate libraries. It minimizes paperwork, reduces manual effort, prevents record duplication, improves inventory tracking, enhances communication through automated notifications, and increases the overall efficiency of library services. Students can easily request and track books, while librarians can manage all library activities through a single cloud-based platform.

Overall, this project demonstrates the practical application of **ServiceNow System Administration** concepts in developing a real-world workflow automation solution. It showcases how ServiceNow can be used to transform traditional library management into a modern, secure, scalable, and efficient digital service that improves productivity and enhances the user experience.

2. IDEATION PHASE

2.1 Problem Statement

Many educational institutions still use manual or semi-automated methods to manage library operations. Maintaining book records, issuing and returning books, tracking due dates, and monitoring book availability require significant manual effort. These traditional methods often lead to delayed services, inaccurate records, misplaced books, duplicate entries, and difficulties in generating reports. As the number of library users and books increases, managing these operations efficiently becomes more challenging. Therefore, there is a need for a cloud-based workflow solution that automates library processes and improves operational efficiency.

2.2 Proposed Solution

The proposed solution is to develop a **Smart Library Workflow** using the **ServiceNow** platform. The application automates the complete library management process, including book requests, approval, issue, return, due-date tracking, and notifications. Students can request books through the Service Portal, while librarians can review requests, issue books, process returns, and update book availability using a centralized system. Automated workflows, Business Rules, Flow Designer, and Notifications ensure that every process is completed accurately and efficiently with minimal manual intervention.

2.3 Objectives

The primary objectives of the Smart Library Workflow are:

- To automate the complete library management process using ServiceNow.
- To reduce manual effort and paperwork in library operations.
- To maintain accurate records of books and library members.
- To simplify the book request, approval, issue, and return process.
- To automatically update book availability after every transaction.
- To send automated email notifications for approvals, due dates, and overdue books.
- To improve efficiency through workflow automation.
- To generate reports and dashboards for effective library monitoring.
- To provide secure role-based access for librarians and students.

2.4 Real-Time Scenarios

Scenario 1: Book Request and Approval

A student logs into the ServiceNow Service Portal and submits a request for a book. The system forwards the request to the librarian for approval. If the book is available, the request is approved, the book is issued, and the student receives an email confirmation along with the due date.

Scenario 2: Book Return Management

After reading the book, the student returns it to the library. The librarian updates the return status in ServiceNow. The workflow automatically changes the book status to **Available**, updates the inventory, and closes the request.

Scenario 3: Overdue Book Notification

If a student does not return a book before the due date, ServiceNow automatically sends reminder notifications to the student. Librarians can also view overdue books in reports and dashboards, enabling timely follow-up and better library management.

3. REQUIREMENT ANALYSIS

Requirement Analysis is one of the most important phases in software development. It involves identifying, analyzing, and documenting the functional and non-functional requirements of the Smart Library Workflow. This phase ensures that the application meets the needs of librarians, students, and administrators while providing a secure, efficient, and user-friendly library management system.

3.1 Functional Requirements

The Smart Library Workflow should provide the following functionalities:

- User authentication and secure login.
- Role-based access for Administrators, Librarians, and Students.
- Book catalog management (Add, Update, Delete, and Search books).
- Student registration and member management.
- Book request submission through the Service Portal.
- Approval workflow for book requests.
- Book issue and return management.
- Automatic update of book availability after issue or return.
- Due date tracking for issued books.
- Email notifications for request approval, issue confirmation, return confirmation, and overdue reminders.

- Fine calculation for overdue books (optional).
- Generation of reports and dashboards for monitoring library activities.

3.2 Non-Functional Requirements

The system should satisfy the following quality requirements:

- **Security:** Protect user data through authentication and role-based access control.
- **Performance:** Process book requests and transactions quickly with minimal response time.
- **Availability:** Ensure the application is accessible whenever required.
- **Reliability:** Maintain accurate and consistent book and member records.
- **Scalability:** Support an increasing number of books, users, and transactions without affecting performance.
- **Usability:** Provide a simple, intuitive, and user-friendly interface.
- **Maintainability:** Allow administrators to modify workflows, forms, and business rules easily.

3.3 Hardware Requirements

The minimum hardware requirements are:

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum (8 GB recommended)
- Storage: 20 GB free disk space
- Internet Connection
- Windows 10/11 or Linux Operating System

3.4 Software Requirements

The Smart Library Workflow is developed using the following software:

- ServiceNow Personal Developer Instance (PDI)
- ServiceNow Studio
- Flow Designer
- Business Rules
- Client Scripts
- UI Policies

- Notifications
- Reports and Dashboards
- Google Chrome or Microsoft Edge Web Browser

3.5 User Requirements

Administrator

- Configure tables, forms, roles, and permissions.
- Manage users and system settings.
- Monitor reports and dashboards.
- Maintain workflows and notifications.

Librarian

- Add and update book information.
- Approve or reject book requests.
- Issue and receive returned books.
- Monitor overdue books.
- Generate library reports.

Student

- Log in to the Service Portal.
- Search available books.
- Submit book requests.
- View issued books and due dates.
- Receive email notifications.
- Return books through the library.

3.6 System Requirements

The Smart Library Workflow requires:

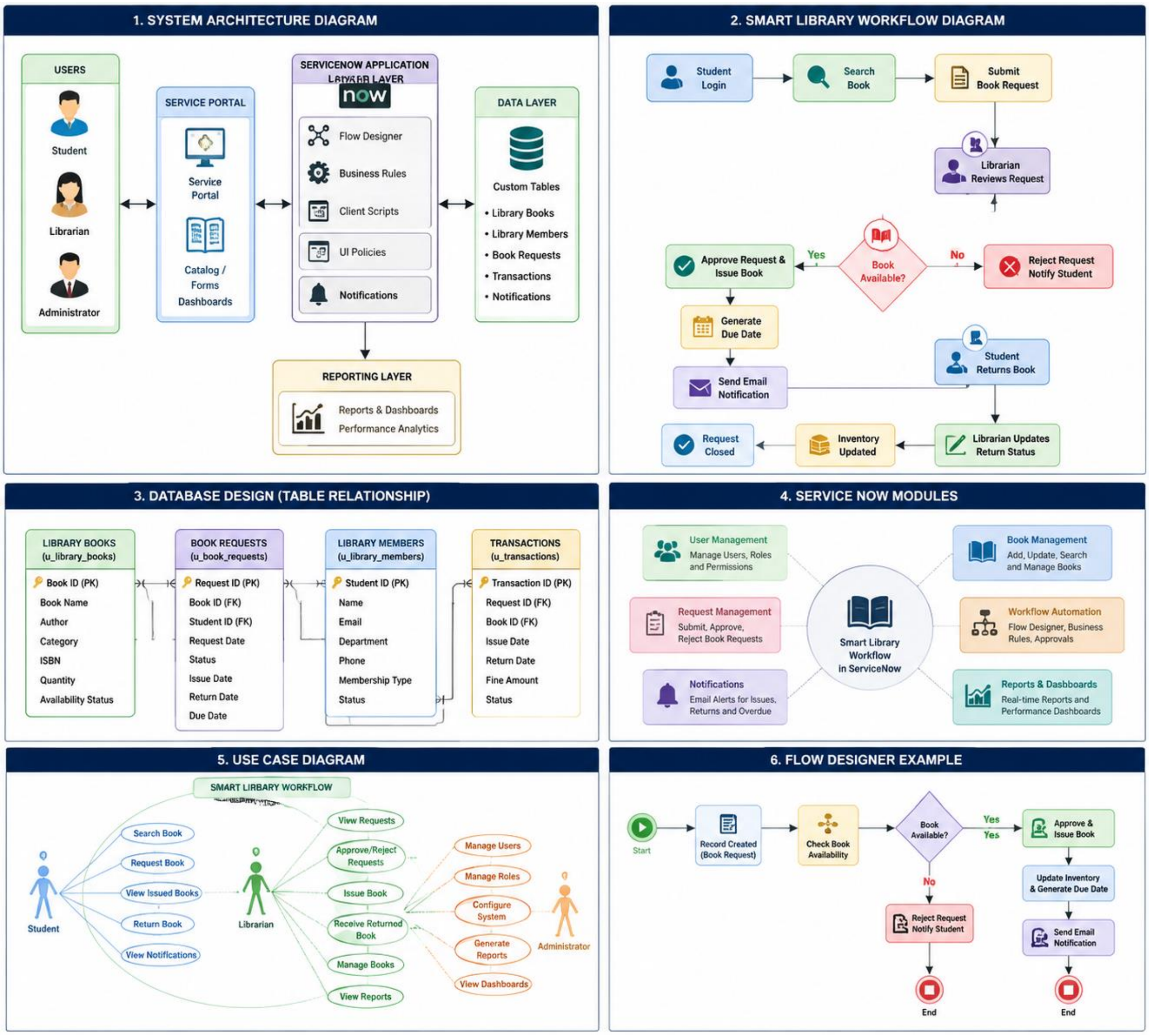
- Stable internet connectivity.
- Active ServiceNow instance.
- Configured email notification service.
- Proper user roles and permissions.

- Database tables for Books, Members, Book Requests, and Transactions.
- Flow Designer for workflow automation.
- Reports and Dashboards for monitoring system activities.

The successful implementation of these requirements ensures that the Smart Library Workflow operates efficiently, reduces manual effort, improves record accuracy, and enhances the overall library management process using the ServiceNow platform.

4. PROJECT DESIGN

The **Project Design** phase defines the overall architecture and working model of the Smart Library Workflow developed using the ServiceNow platform. The system is designed to automate library operations by integrating ServiceNow modules such as custom tables, forms, Flow Designer, Business Rules, Client Scripts, Notifications, Reports, and Dashboards. The application follows a workflow-based architecture that enables smooth communication between students, librarians, and administrators.



4.1 System Architecture

The Smart Library Workflow follows a layered architecture consisting of the following components:

User Layer

- Students access the Service Portal to search and request books.
- Librarians manage book requests, issue books, and process returns.
- Administrators configure the application, manage users, and monitor reports.

Presentation Layer

- Service Portal
- Forms
- Lists

- Catalog Items
- Dashboards

Application Layer

- Flow Designer
- Business Rules
- Client Scripts
- UI Policies
- Notifications

Data Layer

- Custom Tables
- Book Records
- Member Records
- Book Request Records
- Transaction Records

Reporting Layer

- Reports
- Dashboards
- Performance Analytics

The workflow begins when a student submits a book request through the Service Portal. The request is sent to the librarian for approval. If the requested book is available, the librarian approves the request and the system automatically issues the book, updates the inventory, generates a due date, and sends an email notification to the student. When the book is returned, the librarian updates the status, and the system marks the book as available again while closing the request

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4.2 Project Modules

Module 1: User Management

This module manages users and assigns appropriate roles.

Functions:

- User Registration
- Role Assignment
- Login Authentication
- Access Control

Module 2: Book Management

This module maintains the complete book inventory.

Functions:

- Add New Books
- Update Book Details
- Delete Books
- Search Books
- Track Book Availability

Module 3: Member Management

This module stores details of library members.

Functions:

- Student Registration
- Member Information
- Contact Details
- Membership Status

Module 4: Book Request Workflow

Students can request books through the Service Portal.

Workflow:

1. Student submits request.
2. Request is created.
3. Librarian receives notification.

4. Approval process begins.

Module 5: Book Issue Management

After approval, the librarian issues the book.

Functions:

- Issue Book
- Generate Due Date
- Update Book Status
- Send Confirmation Email

Module 6: Book Return Management

This module records returned books.

Functions:

- Accept Returned Books
- Update Inventory
- Close Transaction
- Calculate Fine (Optional)

Module 7: Notification Module

Automated notifications improve communication.

Notifications Include:

- Request Submitted
- Request Approved
- Book Issued
- Due Date Reminder
- Overdue Notification
- Return Confirmation

Module 8: Reports and Dashboards

The system provides real-time monitoring of library activities.

Reports Include:

- Available Books Report
- Issued Books Report
- Overdue Books Report
- Student Activity Report
- Monthly Library Usage Report

4.3 Database Design

The Smart Library Workflow uses the following custom tables:

Table Name	Description
Library Books	Stores book information
Library Members	Stores student/member details
Book Requests	Stores book request records
Book Transactions	Stores issue and return details
Notifications	Stores notification history

4.4 Workflow Design

The workflow of the Smart Library System is:

Student Login
↓
Search Book
↓
Submit Book Request
↓
Librarian Reviews Request
↓
Book Available?

- **Yes → Approve Request → Issue Book → Update Inventory → Send Email → Generate Due Date**
- **No → Reject Request → Notify Student**

↓
Student Returns Book
↓
Inventory Updated
↓
Request Closed

4.5 ServiceNow Components Used

The project is developed using the following ServiceNow components:

- Service Portal
- Custom Tables
- Forms
- Lists
- Flow Designer
- Business Rules
- Client Scripts
- UI Policies
- Email Notifications
- Roles and Permissions
- Reports
- Dashboards
- System Administrator Configuration

The project design ensures that the Smart Library Workflow is secure, scalable, user-friendly, and capable of automating the complete library management process efficiently using the ServiceNow platform.

5. PROJECT PLANNING & SCHEDULING

Project Planning and Scheduling is an essential phase that defines the sequence of activities required to successfully develop and implement the Smart Library Workflow in ServiceNow.

Proper planning ensures that each phase of the project is completed systematically, within the estimated time, and according to the project objectives. The project follows a structured Software Development Life Cycle (SDLC), beginning with requirement analysis and ending with testing and documentation.

5.1 Project Planning

The Smart Library Workflow project was planned to automate library operations by using the ServiceNow platform. The planning process included requirement gathering, designing the system architecture, creating database tables, configuring workflows, implementing automation, testing the application, and preparing project documentation.

The project planning objectives were:

- Identify system requirements.
- Design an efficient workflow for library management.
- Configure ServiceNow tables, forms, and modules.
- Automate library processes using Flow Designer.
- Ensure secure role-based access.
- Test all workflows before deployment.
- Prepare user documentation and project reports.

5.2 Project Schedule

Phase	Activity	Duration
Phase 1	Requirement Analysis	1 Week
Phase 2	System Design & Architecture	1 Week
Phase 3	Creation of Custom Tables & Forms	1 Week
Phase 4	Role Configuration & User Management	3 Days
Phase 5	Workflow Development using Flow Designer	1 Week
Phase 6	Business Rules, Client Scripts & UI Policies	5 Days
Phase 7	Notifications, Reports & Dashboards	4 Days
Phase 8	Functional & Performance Testing	1 Week

Phase	Activity	Duration
Phase 9	Bug Fixing & Optimization	3 Days
Phase 10	Documentation & Final Deployment	1 Week

5.3 Project Timeline

Week	Tasks Completed
Week 1	Requirement gathering and feasibility study
Week 2	System architecture and database design
Week 3	Creation of custom tables, forms, and modules
Week 4	Development of Smart Library Workflow using Flow Designer
Week 5	Implementation of Business Rules, Client Scripts, and Notifications
Week 6	Reports, Dashboards, Testing, Documentation, and Final Deployment

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5.4 Milestones

- ✓ Requirement Analysis Completed
- ✓ System Design Completed
- ✓ Database Tables Created
- ✓ Forms and Modules Configured
- ✓ Flow Designer Workflow Developed
- ✓ Notifications Configured
- ✓ Reports and Dashboards Created
- ✓ Functional Testing Completed
- ✓ Performance Testing Completed
- ✓ Project Documentation Completed

5.5 Deliverables

The final deliverables of the project include:

- Smart Library Workflow Application
- ServiceNow Custom Tables
- Service Portal Interface
- Flow Designer Workflow
- Business Rules
- Client Scripts
- UI Policies
- Email Notifications
- Reports and Dashboards
- User Guide
- Final Project Documentation

The structured planning and scheduling approach ensured that every phase of the Smart Library Workflow project was completed efficiently, resulting in a reliable, secure, and user-friendly library management solution developed on the ServiceNow platform.

6. FUNCTIONAL AND PERFORMANCE TESTING

Functional and Performance Testing is carried out to ensure that the Smart Library Workflow operates correctly, efficiently, and reliably within the ServiceNow platform. Functional testing verifies that each feature performs according to the specified requirements, while performance testing evaluates the system's speed, stability, and responsiveness under normal operating conditions.

6.1 Functional Testing

The following test cases were performed to validate the functionality of the Smart Library Workflow.

Test Case	Expected Result	Status
User Login	User logs in successfully based on assigned role	Passed
Search Book	Available books are displayed correctly	Passed
Submit Book Request	Book request is created successfully	Passed

Test Case	Expected Result	Status
Request Approval	Librarian approves or rejects the request	Passed
Book Issue	Book is issued and due date is generated	Passed
Book Return	Book status is updated to Available	Passed
Email Notification	Student receives email for issue and return	Passed
Due Date Reminder	Reminder notification is sent before due date	Passed
Reports Generation	Reports and dashboards display accurate data	Passed
User Logout	User logs out securely	Passed

Functional Testing Summary

- Student login and authentication worked successfully.
- Book search returned accurate results.
- Book request workflow executed correctly.
- Approval and rejection processes functioned properly.
- Book issue and return records were updated automatically.
- Inventory status changed dynamically after each transaction.
- Notifications were delivered successfully.
- Reports displayed real-time information without errors.

6.2 Performance Testing

Performance testing was conducted to evaluate the system's efficiency and response time under normal usage conditions.

Parameter	Expected Result	Status
Login Response Time	Less than 3 seconds	Passed
Book Search	Instant search results	Passed
Request Processing	Completed within a few seconds	Passed
Flow Designer Execution	Executed without delay	Passed

Parameter	Expected Result	Status
Email Notification	Sent automatically after workflow completion	Passed
Report Generation	Generated within acceptable time	Passed
Multiple User Access	System remained stable	Passed

Performance Testing Results

- The application responded quickly to user requests.
- Flow Designer executed workflows successfully without delays.
- Database records were updated accurately during issue and return operations.
- Reports and dashboards loaded efficiently.
- The system remained stable while handling multiple transactions.
- Automated notifications were generated immediately after workflow execution.

6.3 Testing Tools

The following ServiceNow tools and features were used during testing:

- ServiceNow Personal Developer Instance (PDI)
- Flow Designer Test Execution
- Business Rules
- Client Scripts
- Email Notifications
- Reports and Dashboards
- Service Portal

6.4 Test Outcome

The Smart Library Workflow successfully passed all functional and performance test cases. The application correctly automated book requests, approvals, issue and return management, notifications, and reporting. The system demonstrated reliable performance, secure access, fast response times, and accurate workflow execution, making it suitable for efficient library management using the ServiceNow platform.

7. RESULTS

The **Smart Library Workflow** was successfully developed and implemented using the **ServiceNow platform**. The application effectively automates the complete library management process, including book requests, approvals, book issue, book return, notifications, and reporting. By utilizing ServiceNow features such as Custom Tables, Flow Designer, Business Rules, Client Scripts, UI Policies, Notifications, Reports, and Dashboards, the system minimizes manual effort and improves the overall efficiency of library operations.

The developed workflow provides a user-friendly interface for students, librarians, and administrators. Students can search for available books and submit requests through the Service Portal, while librarians can approve requests, issue books, manage returns, and monitor inventory. The system automatically updates book availability, generates due dates, and sends email notifications at each stage of the workflow.

The application was tested under different scenarios, and all functional modules performed as expected. The workflow executed successfully without errors, and reports generated accurate information regarding book inventory, issued books, overdue books, and member activities.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Automates library management.
- Reduces manual work.
- Easy to use and manage.
- Provides secure role-based access.
- Sends automatic email notifications.
- Generates reports and dashboards.
- Improves accuracy and efficiency.

Disadvantages

- Requires an internet connection.
- Needs a ServiceNow instance.
- Initial setup requires technical knowledge.
- Customization may require scripting skills.
- Regular maintenance is needed.

9. CONCLUSION

The **Smart Library Workflow in ServiceNow** successfully automates the library management process, including book requests, approvals, issue, return, and notifications. The system reduces manual effort, improves accuracy, and provides secure role-based access for users. By using ServiceNow features such as Flow Designer, Business Rules, and Reports, the application offers an efficient and reliable solution for managing library operations. Overall, the project demonstrates how ServiceNow can be used to simplify and improve library management through workflow automation.

10. FUTURE SCOPE

The Smart Library Workflow can be enhanced with additional features to improve its functionality and user experience. Future improvements may include:

- Mobile application integration.
- Barcode or RFID-based book tracking.
- Online book reservation facility.
- AI-based book recommendation system.
- Fine payment through online payment gateways.
- Integration with other educational systems.
- Advanced analytics and dashboards for library management.

11. APPENDIX

Technologies Used

- ServiceNow Platform
- Flow Designer
- Business Rules
- Client Scripts
- UI Policies
- Notifications
- Reports & Dashboards

Modules

- User Management
- Book Management
- Book Request
- Book Issue & Return
- Notifications
- Reports

Input

- Book Details
- Student Details
- Book Request
- Return Request

Output

- Book Issued Successfully
- Book Returned Successfully
- Email Notifications
- Library Reports

- Dashboard

Applications

- Schools
- Colleges
- Universities
- Public Libraries
- Corporate Libraries

Github link:

https://github.com/harshinipotluri/Smart-Library-Request-Workflow-in-ServiceNow

Demo link:

https://github.com/harshinipotluri/Smart-Library-Request-Workflow-in-ServiceNow/blob/main/video_demo.mp4